CS458 Software Verification and Validation, Spring 2025-2026

ARES-X: Integrated Adaptive Survey Ecosystem with TDD and Cross-Platform Orchestration

Group: [Name Surname] - [Email] - [Student ID]

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LLM assistance: OpenAI Codex/GPT-5 in Codex Desktop, used on 2026-04-21 for implementation scaffolding, test design, and report drafting. Final report wording and academic responsibility should be reviewed by the group.

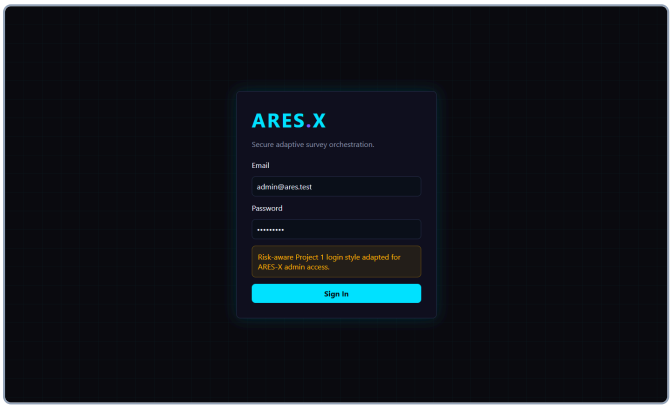
1. Project Overview

ARES-X combines a responsive Web Architect, a JSON-backed Express API, and a native Android Kotlin client. The Web Architect designs and publishes dynamic survey schemas. The mobile client adapts the Project 1 ARES login style and renders the active survey recursively using the same GBCR/RCLR logic.

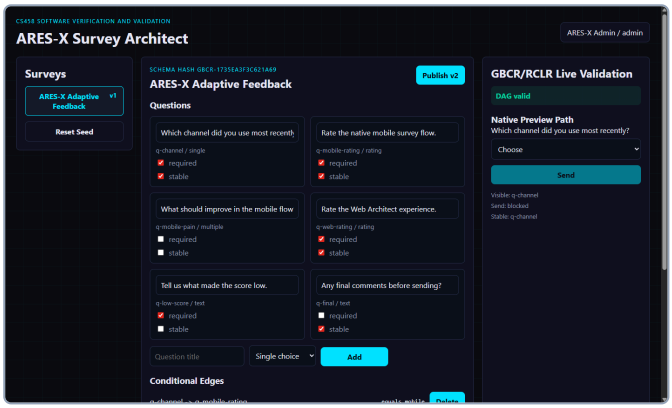
The survey is treated as a directed acyclic graph. Nodes are questions and edges are conditional predicates. The Send button is enabled only when all currently visible required questions on the valid path are answered and no orphan or zombie node exists.

2. Screenshots

Web Login

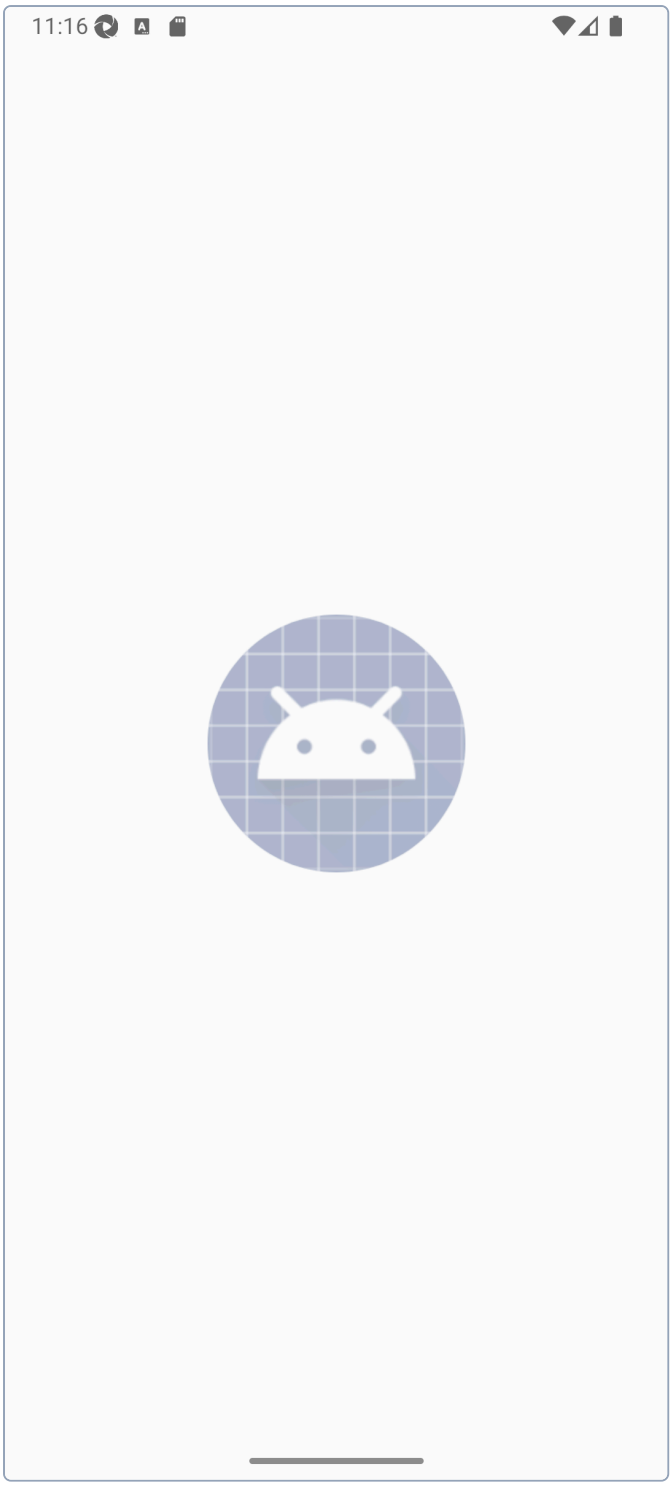


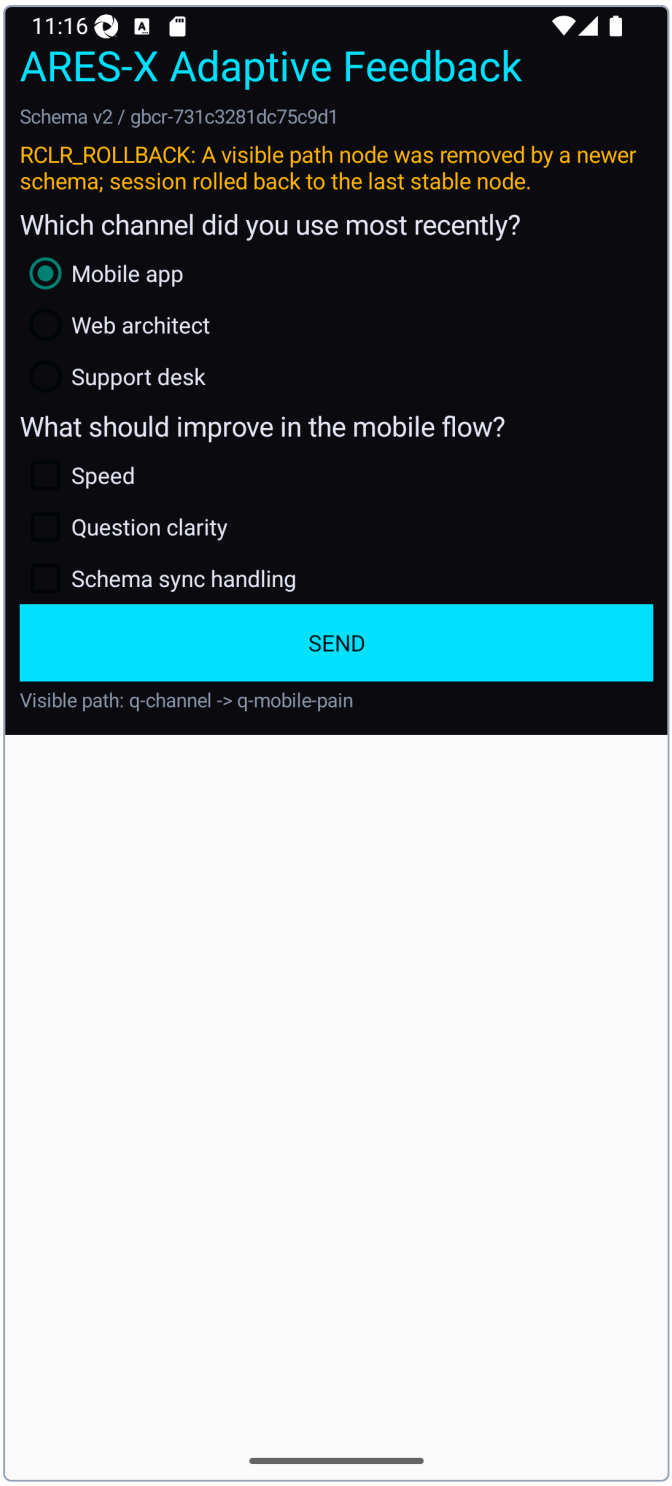
Web Architect



Android Login

Android Conflict Handling


11:16


11:16
ARES-X Adaptive Feedback
Schema v2 / gbcr-731c3281dc75c9d1
RCLR_ROLLBACK: A visible path node was removed by a newer schema; session rolled back to the last stable node.
Which channel did you use most recently?
Mobile app
Web architect
Support desk
What should improve in the mobile flow?
Speed
Question clarity
Schema sync handling
SEND
Visible path: q-channel -> q-mobile-pain

3. UML Documentation

Use-case Diagram

```
Admin -> Login, Design Survey Schema, Publish Versioned DAG
Mobile Participant -> Native Login, Answer Dynamic Survey, Submit Valid Path
Publish -> Cross-Platform Schema Sync -> Atomic Recovery / Rollback
```

Activity Diagram

```
Start -> Login -> Fetch Survey -> Render Roots -> Answer -> Run RCLR
Run RCLR -> Version Changed? -> Atomic Recovery or Rollback
Run RCLR -> Complete Visible Required Path? -> Enable Send -> Submitted
```

State Diagram

```
LoggedOut -> SurveyList -> ActiveSession -> Resolving
Resolving -> ActiveSession | AtomicRecovered | RolledBack | ConflictFlagged
ActiveSession -> Submitted
```

Sequence Diagram

```
Admin Web -> Backend: publish schema v2
Android -> Backend: PATCH answers(clientVersion=v1)
Backend -> RCLR: compare v1 DAG with current DAG
RCLR -> Backend: atomic_recovery or RCLR_ROLLBACK
Backend -> Android: sanitized answers, stable node, current schema
```

Class Diagram

```
SurveySchema *-- SurveyQuestion
SurveySchema *-- ConditionalEdge
RclrEngine --> SurveySchema
JsonStore --> SurveySchema
MainActivity --> RclrEngine
```

4. GBCR/RCLR Algorithm

GBCR validates the schema graph before publish. It checks duplicate nodes, missing edge endpoints, missing predicate references, duplicate edges, root availability, and cycles. RCLR starts at root questions, traverses only satisfied conditional edges, and records visible nodes in topological order.

```
export function resolveVisibility(schema, answers) {
  const roots = validation.roots;
  visit(root);
  for each outgoing edge:
    if predicateSatisfied(edge, answers) visit(edge.to)
  blockers = visible required questions without answers
  sendEnabled = blockers.empty && no orphanQuestionIds && DAG valid
}
```

When mobile sends answers for an older schema version, the backend compares the old visible path with the current DAG. If answers map cleanly, it returns `atomic_recovery`. If a visible node was removed, an answered node was deleted, or a zombie/orphan would appear, it returns `RCLR_ROLLBACK` or `RCLR_CONFLICT` and the mobile UI re-renders from the stable node.

5. Verification Strategy

Layer	Automation	Purpose
Shared logic	Vitest	DAG validation, visibility, Send gating, atomic recovery, rollback.
Backend	Supertest	Login, publish version increment, session conflict responses.
Android	JUnit	Kotlin RCLR parity and hidden answer clearing.
Web	Selenium	Login, designer edit, validation status, publish reflection.
Mobile	Appium	10 non-trivial native UI and logic cases.
Cross-platform	Selenium + Appium	Concurrent schema mutation during mobile answer selection.

Important Test Excerpts

```
expect(resolveSyncConflict(oldSchema, newSchema, answers).action).toBe('rollback');
assert.match(await appium.text('conflict-banner'), /RCLR_ROLLBACK|RCLR_CONFLICT/);
assert.doesNotMatch(await appium.text('visible-path'), /undefined|zombie/i);
```

6. Test Users and Commands

User	Password	Role
admin@ares.test	Admin123!	Admin
alice@ares.test	Test1234!	Mobile participant
bob@ares.test	Secure99#	Mobile participant
carol@ares.test	Pass@word1	Mobile participant

```
npm test
npm run typecheck
npm run build:web
gradle -p mobile clean testDebugUnitTest
gradle -p mobile assembleDebug
npm run test:e2e:web
npm run test:e2e:mobile
npm run test:e2e:sync
```